# IMO Training – Number Theory
# Order, Valuation and Diophantine Equations

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## Notes

Unless otherwise specified, let $p$ be a prime number.

**Theorem.** (Fermat's Little Theorem) $\forall a \in \mathbb{Z}, a^p \equiv a \pmod{p}$. Equivalently, $\forall a \in \mathbb{Z}$ with $(a,p) = 1$, $a^{p-1} \equiv 1 \pmod{p}$.

**Definition.** (Euler Totient Function) $\forall n \in \mathbb{N}$, $\varphi(n) = \#\{m \in \mathbb{N} \mid 1 \leq m < n, (m,n) = 1\}$. Equivalently, for $n = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k}$, where $p_1, p_2, \dots, p_k$ being distinct primes, and integers $a_1, a_2, \dots, a_k \geq 1$, $\varphi(n) = n\left(1 - \dfrac{1}{p_1}\right)\left(1 - \dfrac{1}{p_2}\right)\dots\left(1 - \dfrac{1}{p_k}\right)$.

**Theorem.** (Fermat-Euler Theorem) $\forall a \in \mathbb{Z}, n \in \mathbb{N}$ s.t. $(a,n) = 1$, $a^{\varphi(n)} \equiv 1 \pmod{n}$.

**Theorem.** (Wilson's Theorem) $p$ is a prime iff $(p-1)! \equiv -1 \pmod{p}$.

**Definition.** (Order and Primitive Roots) For $a \in \mathbb{Z}, n \in \mathbb{N}$ with $(a,n) = 1$, the *order of $a$ modulo $n$* ($ord_n(a)$) is defined by the minimal $m \in \mathbb{N}$ s.t. $a^m \equiv 1 \pmod{n}$. Furthermore, if $ord_n(a) = \varphi(n)$, then $a$ is a *primitive root modulo $n$*.

**Definition.** (Valuation) For $m \in \mathbb{N}$, the *$m$-adic valuation* of $n$ is the $a \in \mathbb{N}$ s.t. $m^a \mid n$ and $m^{a+1} \nmid n$.

**Example.** Prove that if $a, m, n \in \mathbb{N}$, where $a, m$ are coprime, then $a^n \equiv 1 \pmod{m}$ iff $\operatorname{ord}_m(a) \mid n$.

**Example.** Prove that for any prime $p$, every prime divisor of $2^p - 1$ is greater than $p$.

**Example.** Let $p$ be an odd prime, and let $q, r$ be primes s.t. $p \mid q^r + 1$. Prove that $2r \mid p - 1$ or $p \mid q^2 - 1$.

**Example.** Let $a, n \in \mathbb{N}$ with $a > 1$. If $p$ is an odd prime s.t. $p \mid a^{2^n} + 1$, prove that $2^{n+1} \mid p - 1$.

**Example.** (Putnam 1972)Let $n \in \mathbb{Z}, n \geq 2$. Prove that $n \nmid 2^n - 1$.

**Example.** Prove that for all $a, n \in \mathbb{N}$ with $a > 1$, $n \mid \varphi(a^n - 1)$.

**Example.** Find all $x, y \in \mathbb{Z}$ s.t. $3x + 4y = 1$.

**Example.** Find all $x, y \in \mathbb{Z}$ s.t. $11x + 22y = 13$.

**Example.** Find all $n \in \mathbb{N}$ s.t. $2^8 + 2^{11} + 2^n$ is a perfect square.

**Example.** Find all $x, y \in \mathbb{Z}$ s.t. $x^3 + y^3 = 2006$.

**Example.** Find all $x, y \in \mathbb{Z}$ s.t. $y^2 = x^3 + 7$.

**Example.** Find all $x, y \in \mathbb{Z}_{\geq 0}$ s.t. $1 + 3^x = 2^y$.

**Example.** Find all $x, y \in \mathbb{Z}_{\geq 0}$ s.t. $1 + 2^x = 3^y$.

**Example.** Find all $x, y, z \in \mathbb{N}$ s.t. $3^x + 4^y = 5^z$.

# Exercises

1. (AIME 2001) How many positive integer multiples of 1001 can be expressed in the form $10^j - 10^i$, where $i$ and $j$ are integers and $0 \leq i < j \leq 99$?

2. (IMO 1978) Let $m$ and $n$ be natural numbers such that $1 \leq m < n$. In their decimal representations, the last three digits of $1978^m$ are equal, respectively, to the last three digits of $1978^n$. Find $m$ and $n$ such that $m + n$ has its least value.

3. (IMO 1990) Find all positive integers $n > 1$ such that $\dfrac{2^n + 1}{n^2}$ is an integer.

4. (Bulgaria 1996) Find all primes $p, q$ s.t. $pq \mid (5^p - 2^p)(5^q - 2^q)$.

5. (USA TST 2003) Find all ordered prime triples $(p, q, r)$ s.t. $p \mid q^r + 1$, $q \mid r^p + 1$, $r \mid p^q + 1$.

6. (China 2009) Find all primes $p, q$ s.t. $pq \mid 5^p + 5^q$.

7. (China 2005) Find all $x, y, z, w \in \mathbb{Z}_{\geq 0}$ s.t. $2^x 3^y - 5^z 7^w = 1$.

8. (Bulgaria 2006) Find all $t, x, y, z \in \mathbb{N}$ s.t. $2^t = 3^x 5^y + 7^z$.

9. Find all $x, y \in \mathbb{Z}$ s.t. $x^3 - y^3 = xy + 61$.

10. Find all $x, y \in \mathbb{Z}$ s.t. $x^2 - y! = 2001$.

11. For $m, n \in \mathbb{N}$, find the minimum value of $|12^m - 5^n|$.

12. Find all $x, y \in \mathbb{Z}$ s.t. $x^3 + y^4 = 19^{19}$.

13. Find all $k \in \mathbb{N}$ s.t. $\exists\, a, b \in \mathbb{N}$ with $a^2 + b^2 = kab$.

14. If $a, b \in \mathbb{N}$ and $b^2 + ab + 1 \mid a^2 + ab + 1$, prove that $a = b$.

15. Show that for $a, b, c \in \mathbb{N}$ with $a^2 + b^2 = c^2$, $\exists\, m, n \in \mathbb{Z}$ s.t. $(a, b, c) = (m^2 - n^2, 2mn, m^2 + n^2)$.

16. Find all $x, y \in \mathbb{Z}$ s.t. $x^2 - 2y^2 = 1$.

17. (Bulgaria 1998) Let $m, n \in \mathbb{N}$ such that $A = \dfrac{(m+3)^n + 1}{3m} \in \mathbb{Z}$. Prove that $A$ is odd.

18. (IMO 1996) Let $a, b \in \mathbb{N}$ s.t. $15a + 16b, 16a - 15b$ are both perfect squares. Find the minimum value of the smaller square.